

Muhammad Shaheer Luqman

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AI researcher working in reinforcement learning, computer vision, and robot manipulation. Published across arXiv (2×) and IEEE, ICPC Gold Medalist, and summa cum laude graduate (3.90/4.00).

Education

National University of Computer and Emerging Sciences (FAST)

Karachi, Pakistan

BS COMPUTER SCIENCE

Sep 2021 – Jun 2025

- CGPA: **3.90 / 4.00** (Ranked **9 / 440**)
- Summa Cum Laude Honors

Experience

Retrocausal

Remote (US-based)

APPLIED RESEARCH ENGINEER

Jun 2025 – Present

- Research on robot manipulation and Vision-Language-Action systems, unsupervised action segmentation, and temporal long-video understanding — **two papers submitted** (see [Research & Publications](#)).
- Built and shipped end-to-end AI systems for industrial clients (incl. **BMW**), from model training to GPU deployment on **AWS** and **RunPod**.

Reinforcement Learning Robotics VLMs Computer Vision PyTorch

Retrocausal

Remote (US-based)

SOFTWARE ENGINEER INTERN

Mar 2024 – Jun 2024

- Built a video activity-recognition pipeline with **body pose estimation** and **vision-language models** for automated industrial workflow analysis.
- Developed **LLM prompt-engineering** systems for analysis of industrial process data.

Computer Vision VLMs LLMs MediaPipe Python

Research & Publications

Enhancing Human-Likeness in Reinforcement Learning Agents via Hierarchical Macro Action Quantization

Under Review

arXiv:2605.30928

A human-like RL framework (HiMAQ) that encodes human demonstrations into a two-level hierarchy of macro actions for dexterous humanoid-hand control. On D4RL it improves human-likeness over the non-hierarchical baseline while matching or exceeding success rates across IQL, SAC, and RLPD. [View on arXiv](#)

Robot Manipulation Hierarchical Action Quantization Reinforcement Learning Imitation Learning

Unsupervised Skeleton-Based Action Segmentation via Hierarchical Spatiotemporal Vector Quantization

Under Review

arXiv:2604.15196

A hierarchical spatiotemporal vector-quantization framework for unsupervised skeleton-based action segmentation, achieving state-of-the-art on the HuGaDB, LARa, and BABEL benchmarks. [View on arXiv](#)

Unsupervised Learning Action Segmentation Vector Quantization Skeleton-Based Recognition

Ensuring Child Safety: An IoT-Based Surveillance System for Remote Monitoring and Detection of Anomalous Behavior

ICETST 2024

IEEE Xplore

An IoT surveillance system pairing hardware sensors with AI-driven anomaly detection for real-time child-safety alerts. [View on IEEE Xplore](#)

IoT Systems Anomaly Detection Surveillance Embedded Systems

Research Interests: Robot Manipulation, Computer Vision, Reinforcement Learning, Vision-Language Models, IoT + AI Systems

Certifications

IELTS Academic — English Proficiency

Jul 2026

Overall Band: **7.5 / 9** • Reading **9.0** • Listening **8.5**

Technical Skills

Languages Python, C / C++, JavaScript.

ML Frameworks PyTorch, Hugging Face, Scikit-learn; YOLO, MediaPipe; Stable Diffusion; BERT, Whisper.

Software Frameworks Django, FastAPI, Flask; Express.js, React, Next.js, Vue.

Developer Tools AWS, RunPod; Docker, Git, Linux.

Projects

PixelForge — Generative AI Marketing Platform

Final Year Project ([GitHub](#))

Full-stack generative pipeline combining **BLIP** captioning, **Gemini** text generation, and diffusion-based image and video synthesis for product-aware marketing content.

LushWear — Business Operation Management

Client ([GitHub](#))

Production-ready system, built, sold, and actively maintained for a local business to run its day-to-day operations. Full-stack inventory, order, and storefront management with a **FastAPI** backend, **Electron** desktop app, **Supabase** real-time database, and two-way **Shopify** integration.

More on my [portfolio](#) and [GitHub](#).

Honors & Awards

- **ICPC Gold Medalist** — ICPC National Competitive Programming (*Nov 2024*)
- **Summa Cum Laude** — Graduated with a high GPA (3.90 / 4.00), National University of Computer and Emerging Sciences (FAST) (*2025*)
- **ICPC Bronze Medalist** — ICPC National Competitive Programming (*Feb 2024*)
- **Winner** — Code Sprint, PROCOM 2023 (*2023*)
- **Best Team Award** — Coder's Cup, ACM FAST
- **4x Rector's List (4.0 GPA)** — NUCES (FAST), 4 semesters
- **4x Dean's List (3.5+ GPA)** — NUCES (FAST), 4 semesters